Endpoint Security Baselines

Implementation Solution Guide

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Executive Summary — Endpoint Security Baselines (Microsoft Intune)

White-label · Intune Solutions

Hardening Windows by hand means choosing among hundreds of security settings — and getting some wrong. **Security Baselines** in Microsoft Intune apply Microsoft's own recommended, pre-configured set of hardening settings in one move, giving you an expert-grade secure posture without the guesswork, then letting you customize and keep it current as Microsoft updates the baseline.

This is a complete, **deploy-it-today** package for implementing Endpoint Security Baselines with Microsoft Intune, written for an administrator with no prior baseline experience — including how to detect and resolve the setting conflicts baselines can introduce.

What you get

- A plain-language **Solution Overview** of what baselines are and how they apply.
- A **Requirements Guide** with verify / implement-if-missing for every prerequisite.
- An **Implementation Guide** of precise, click-by-click Intune steps in the required order.
- Five **PowerShell scripts** for posture discovery, baseline validation, tenant reporting, conflict checking, and remediation.
- **Testing** and **Troubleshooting** guides for confident rollout.

What this delivers

A deployed **Security Baseline for Windows** (and optionally the **Microsoft Defender for Endpoint** and **Microsoft Edge** baselines), assigned to a pilot, with a clear process to **review per-setting status, detect conflicts, and version-update** the baseline over time.

The standard deployment model

Assess → **Configure** → **Pilot** → **Validate** → **Expand.**

Outcome

An expert, consistent, Microsoft-recommended hardening posture across the fleet — without manual setting-by-setting work — deployable by a single administrator and validated before rollout.

1. Solution Overview — Endpoint Security Baselines

What Security Baselines are

A security baseline in Microsoft Intune is a pre-configured group of security settings, set to the values Microsoft's security team recommends. Instead of configuring hundreds of individual policy settings, you deploy a baseline and the device adopts a vetted, hardened configuration — which you can then

customize and keep current as Microsoft publishes new baseline versions.

Available baselines

- **Security Baseline for Windows 10 and later** — the core OS hardening baseline.
- **Microsoft Defender for Endpoint Baseline** — security settings for MDE-managed devices.
- **Microsoft Edge Baseline** — browser security settings.

How it works

1. You create a **profile from a baseline version**; it starts at Microsoft's recommended values.
2. You review and, where needed, **adjust** specific settings for your environment.
3. You **assign** the profile to a pilot group; Intune applies the settings.
4. Intune reports **per-setting status** and flags **conflicts** with other profiles.
5. When Microsoft releases a **new baseline version**, you update the profile to stay current.

Benefits

- **Expert hardening, fast** — Microsoft's recommendations applied in one move.
- **Consistency** — every device adopts the same vetted posture; no drift.
- **Customizable** — adjust the few settings your environment requires.
- **Maintained** — version updates track Microsoft's evolving recommendations.

Limitations (be honest with buyers)

- Baselines can **conflict** with existing configuration profiles/GPOs — conflict detection and resolution are essential (covered here).
- Some hardened settings may affect line-of-business apps — pilot before fleet.
- Baselines are **versioned**; staying current requires periodic profile updates.
- A baseline is a strong starting point, not a substitute for the specific solutions (BitLocker, ASR, Firewall) where you want explicit control.

The standard deployment model

> ***Pick the latest baseline version → create a profile → review/adjust → assign to a pilot → resolve conflicts → expand → keep current.*** *repeatable pattern:* ***Assess → Configure → Pilot → Validate → Expand.***

2. Requirements Guide — Endpoint Security Baselines

Verify **every** requirement before configuring. Each entry: **Description · Why · Verify · Implement-if-missing.** Category-wide prerequisites are in

`global-requirements-intune-solutions.md`.

2.1 Licensing

Description: Intune (Plan 1). The Defender for Endpoint baseline needs MDE licensing. **Why:** Intune delivers baselines; the MDE baseline only applies to MDE-licensed/onboarded devices. **Verify:** M365 admin center → **Billing** → **Licenses**. **Implement:** Assign M365 E3/E5 or Business Premium; add MDE licensing for the Defender baseline.

2.2 Device enrollment

Description: Targeted devices enrolled in Intune. **Why:** Baselines apply only to enrolled devices. **Verify:** Intune → **Devices** → **All devices**. **Implement:** Enable automatic enrollment; enroll devices.

2.3 Supported Windows OS

Description: Windows 10/11 current build. **Why:** Baseline settings map to OS capabilities; older builds may not honor all settings. **Verify:** `winver`; run `ESB-Discovery.ps1` for the current posture. **Implement:** Patch/upgrade to a supported build.

2.4 Inventory of existing configuration

Description: Awareness of existing config profiles, security policies, and any GPOs. **Why:** Baselines can conflict with settings you already manage elsewhere. **Verify:** Intune → **Devices** → **Configuration**; **Endpoint security** policies; run `ESB-ConflictCheck.ps1` after assignment. **Implement:** Document current policies; plan to consolidate or scope to avoid conflicts.

2.5 (Optional) Defender for Endpoint onboarding

Description: Devices onboarded to MDE. **Why:** Required for the Microsoft Defender for Endpoint baseline. **Verify:** Intune → **Endpoint security** → **Microsoft Defender for Endpoint**. **Implement:** Deploy the **Defender for Endpoint Integration** solution.

2.6 Pilot group

Description: A pilot security group. **Why:** Baselines change many settings — pilot to catch app/usability impact. **Verify:** Entra → **Groups**. **Implement:** Create a `Baseline-Pilot` security group.

Checklist

- Intune licensing assigned (MDE licensing for the Defender baseline)
- Devices enrolled
- Supported Windows build — `ESB-Discovery.ps1`
- Existing config/GPO inventory documented
- (Optional) Devices onboarded to MDE for the Defender baseline

- Pilot Entra group created

Global Requirements — Intune Solutions (category-level)

These requirements apply to **every** solution in the **Intune Solutions** category. They are published once at the category level so individual listings stay small. Confirm these before deploying *any* Intune solution. Each entry: **Description · Why · Verify · Implement-if-missing.**

G1. Microsoft Entra ID tenant

Description: An active Microsoft Entra ID (Azure AD) tenant. **Why:** It is the identity backbone for every cloud and Intune-managed resource. **Verify:** Sign in to `https://entra.microsoft.com` as an administrator. **Implement:** Create a Microsoft 365/Azure subscription, which provisions a tenant.

G2. Microsoft Intune subscription

Description: An Intune-bearing license plan. **Why:** Intune delivers all configuration, compliance, and security policies. **Verify:** `https://intune.microsoft.com` loads; **Tenant administration** → **Tenant status** shows active. **Implement:** Assign Microsoft 365 E3/E5, Business Premium, or standalone Intune Plan 1.

G3. Administrative roles

Description: Appropriate Entra/Intune admin roles for the operator. **Why:** Creating policies and enrollment settings requires elevated rights. **Verify:** Entra → **Roles and administrators**; confirm **Intune Administrator**. **Implement:** Assign **Intune Administrator**; reserve Global Administrator for tenant-wide changes.

G4. Device enrollment in Intune

Description: Targeted Windows devices are enrolled and manageable. **Why:** Policies only reach enrolled devices. **Verify:** Intune → **Devices** → **All devices**; the device appears and is **compliant**. **Implement:** Enable **Automatic Enrollment** and join devices to Entra ID.

G5. Pilot security group

Description: A Microsoft Entra security group for pilot/targeted users or devices. **Why:** Every solution is piloted before fleet rollout. **Verify:** Entra → **Groups**; the pilot group exists. **Implement:** Entra → **Groups** → **New group** → **Security** → add members.

G6. Supported Windows baseline

Description: Windows 10 21H2+ or Windows 11 with current updates. **Why:** Modern Intune features assume a patched, supported OS. **Verify:** `winver`, or Intune → device → **Hardware**. **Implement:** Deploy **Update rings** or upgrade to Windows 11.

G7. Network access to Microsoft cloud endpoints

Description: Outbound HTTPS to Microsoft 365, Entra, and Intune service URLs. **Why:** Enrollment, policy sync, and authentication depend on these endpoints. **Verify:** From a device, reach `https://login.microsoftonline.com` and `https://enrollment.manage.microsoft.com`. **Implement:** Allow Microsoft's published Office 365/Intune URL & IP ranges through proxy/firewall.

The standard deployment model (all solutions): Assess → Configure → Pilot → Validate → Expand.

3. Implementation Guide — Endpoint Security Baselines

Follow these steps **in order**, in the **Microsoft Intune admin center** (`https://intune.microsoft.com`).

> ***Model:*** *Deploy the latest Windows security baseline to a pilot, resolve conflicts, then expand and keep it version-current.*

Step 1 — Create the pilot group

1. Entra → **Groups** → **New group** → Security → `Baseline-Pilot` → add pilot devices. **Create**.

Step 2 — Review available baselines

1. Intune → **Endpoint security** → **Security baselines**.
2. Note the available baselines and their **latest version** (e.g. **Security Baseline for Windows 10 and later**).

Step 3 — Create a baseline profile

1. Select **Security Baseline for Windows 10 and later** → **Create profile**.
2. **Name:** `Baseline - Windows - Pilot`. **Next**.
3. Confirm the profile is created from the **latest version** shown.

Step 4 — Review and adjust settings

1. On **Configuration settings**, the categories show Microsoft's recommended values.
2. Leave defaults unless a setting conflicts with a required app or another solution you manage explicitly (e.g. if you manage BitLocker/Firewall/ASR via their own solutions, decide which owns those settings to avoid conflicts).
3. Click **Next**.

Step 5 — Scope tags (optional)

1. Add scope tags if used, else **Next**.

Step 6 — Assign to the pilot group

1. **Assignments** → **Add groups** → `Baseline-Pilot`. **Next** → **Create**.

Step 7 — Monitor application and conflicts

1. Intune → **Endpoint security** → **Security baselines** → profile → **View report** / **Per-setting status**.
2. Resolve **Conflicts** — a setting controlled by two policies. Run `ESB-ConflictCheck.ps1` to list overlapping policies; make one policy authoritative per setting.

Step 8 — Validate the pilot

1. On a pilot device, run `ESB-Validation.ps1` → confirms representative baseline settings applied.
2. Run `ESB-TenantReport.ps1` → confirms assignment + per-setting success across the pilot. Complete the **Testing Guide**.

Step 9 — Expand and keep current

1. Add a production group to the baseline **Assignments**; roll out in waves.
2. When Microsoft releases a **new baseline version**, select the profile → **Change version** → review changes → update.
3. Re-validate with `ESB-Validation.ps1` / `ESB-TenantReport.ps1`.

Order of operations

```
Pilot group → Review baselines → Create profile (latest version) → Adjust → Assign → Resolve conflicts → Validate → Expand → Version-update
```

5. Testing Guide — Endpoint Security Baselines

Run after the pilot baseline applies (Implementation Step 8) and after each production wave or version update.

Test 1 — Profile assigned and applying

Do: Intune → **Endpoint security** → **Security baselines** → profile → **Device status**. **Expected:** Pilot devices show **Succeeded** (not Error).

Test 2 — No unresolved conflicts

Do: Review **Per-setting status** for **Conflict**; run `ESB-ConflictCheck.ps1`. **Expected:** No settings in **Conflict** (each contested setting has one authoritative owner).

Test 3 — Representative settings applied

Do: On a pilot device run `ESB-Validation.ps1` (checks a sample of hardened settings). **Expected:** Sampled baseline settings are at the expected hardened values.

Test 4 — No critical app/usability breakage

Do: Have pilot users run their core line-of-business apps for a day or two. **Expected:** No baseline-induced breakage; capture and adjust any setting that blocks a required app.

Test 5 — Compliance intact

Do: Intune → **Devices** → **Compliance**. **Expected:** Devices remain **Compliant**.

Test 6 — Version update flows cleanly

Do: When testing a baseline version change, review the **change summary** before applying. **Expected:** You can see what changed and apply it without unexpected conflicts.

Test 7 — Tenant report

Do: Run `ESB-TenantReport.ps1`. **Expected:** Assignment and per-setting success summary for the pilot group.

Pass/fail gate

Gate	Requirement
Pilot pass	Tests 1–4 pass; conflicts resolved; no critical app breakage
Wave pass	Tests 1–2, 5, 7 pass before the next wave

6. Troubleshooting Guide — Endpoint Security Baselines

Use `ESB-Validation.ps1` / `ESB-Discovery.ps1` for device posture and `ESB-ConflictCheck.ps1` for overlaps, `ESB-Remediation.ps1` to force a sync.

A. Settings in conflict (most common)

Symptom	Cause	Remediation
Per-setting status shows **Conflict**	Two policies set the same setting differently	Run `ESB-ConflictCheck.ps1`; pick one authoritative policy for that setting; remove/scope the other
Baseline fights a dedicated solution	BitLocker/Firewall/ASR managed both by baseline and its own solution	Decide one owner per area; exclude that area from the baseline or from the dedicated policy
Legacy GPO overrides baseline	On-prem GPO sets the same control	Disable/scope the GPO away from Intune-managed devices

B. Baseline not applying

Symptom	Cause	Remediation
Profile "Error"/"Pending"	Device not synced / not enrolled	Run `ESB-Remediation.ps1` (forces sync); confirm device in `Baseline-Pilot`
Some settings "Not applicable"	OS build doesn't support the setting	Patch/upgrade Windows; expected for down-level builds

C. App or usability breakage

- A hardened setting blocks a required app → identify it via per-setting status / event logs, then **adjust that single setting** in the baseline profile (don't abandon the whole baseline).
- Pilot exists precisely to catch this before fleet rollout.

D. Version update problems

- Unexpected changes after updating a baseline version → review the **change summary** first; test the new version on the pilot before production.
- A setting reverted → a newer version may have changed a default; re-apply your customization.

E. Diagnostics quick reference

```
# sample of hardened values the baseline sets:
Get-MpPreference | Select-Object PUAProtection, EnableNetworkProtection
(Get-NetFirewallProfile).Enabled
Get-ItemProperty 'HKLM:\SYSTEM\CurrentControlSet\Control\Lsa' -Name LimitBlankPasswordUse -EA SilentlyCont
```

Intune → **Endpoint security** → **Security baselines** → **(profile)** → **Per-setting status** is the authoritative view of what applied and what conflicted.

F. When to escalate

If a setting shows **Error** (not Conflict) on a healthy, synced, supported device and persists after remediation, capture the per-setting detail and open a Microsoft support case.

7. Best Practices — Endpoint Security Baselines

- **Always pilot.** Baselines change many settings at once; pilot to catch app/usability impact before fleet rollout.
- **Start from the latest version.** Create profiles from the newest baseline so you begin at current recommendations.
- **One owner per setting.** Decide whether the baseline or a dedicated solution (BitLocker, Firewall, ASR) owns a given control — never both — to avoid conflicts.

• **Resolve conflicts immediately.** A setting in Conflict is effectively unmanaged; use `ESB-ConflictCheck.ps1` and consolidate.

• **Change settings surgically.** When a baseline setting breaks an app, adjust that one setting — don't discard the whole baseline.

• **Retire overlapping GPOs.** Don't run a legacy GPO and an Intune baseline against the same control.

• **Track versions.** When Microsoft ships a new baseline version, review the change summary, pilot it, then update.

• **Layer, don't duplicate.** Use the baseline for broad hardening and the dedicated solutions for areas you want explicit, reportable control over.

• **Document deviations.** Record every setting you changed from Microsoft's recommendation and why — for audits and future updates.

• **Monitor continuously.** Use `ESB-TenantReport.ps1` to keep per-setting success high across the fleet.

8. Deliverables Summary — Endpoint Security Baselines

Documentation

Document	Purpose
Executive Summary	Business case and what's in the box
1. Solution Overview	What baselines are, available baselines, how they apply, benefits, limitations
2. Requirements Guide	Every prerequisite with Verify / Implement-if-missing
Global Requirements (Intune Solutions)	Category-wide prerequisites, published once
3. Implementation Guide	Exact, ordered Intune steps including conflict resolution and version updates
5. Testing Guide	Validation procedures with pass/fail gates
6. Troubleshooting Guide	Conflicts and field issues
7. Best Practices	Field-proven recommendations

PowerShell scripts

Script	What it does
`ESB-Discovery.ps1`	Snapshot of the device's current security posture (sample of baseline-relevant settings)
`ESB-Validation.ps1`	Confirms representative baseline settings are applied at hardened values

Script	What it does
`ESB-TenantReport.ps1`	Tenant-wide baseline assignment + per-setting state (Microsoft Graph)
`ESB-ConflictCheck.ps1`	Lists security/config policies that may conflict with the baseline (Microsoft Graph)
`ESB-Remediation.ps1`	Forces an MDM sync so the baseline re-applies

What the buyer receives

- This implementation guide as a branded **Word** document and **PDF**.
- All five **PowerShell scripts** in a single ZIP.
- A repeatable process following the repeatable model: **Assess** → **Configure** → **Pilot** → **Validate** → **Expand.**

Outcome

An expert, consistent, Microsoft-recommended hardening posture across the fleet — without manual setting-by-setting work.

Appendix A — PowerShell Scripts

Included in the accompanying ZIP. Each script's header documents full usage.

ESB-ConflictCheck.ps1

ESB-Discovery.ps1

ESB-Remediation.ps1

ESB-TenantReport.ps1

ESB-Validation.ps1